Chapter 15, Problem 4PP

Problem

Find the Laplace transform of *f*(*t*) = *t*2 cos 3*t* *u*(*t*).

Step-by-step solution

Step 1 of 5

Refer to equation 15.34 in the textbook.

Consider the given function.

$$f(t) = t^2 \cos(3t) u(t)$$

Consider the Laplace transform pair.

$$\mathrm{L}\left[\cos(at)\right] \leftrightarrow \left(\frac{s}{s^2 + a^2}\right)$$

Obtain the Laplace transform.

$$\mathrm{L}\left[\cos(3t)\right] = \frac{s}{s^2 + 3^2}$$

Step 2 of 5

Determine the Laplace transform of $f(t)$.

$$F(s) = \mathrm{L}\left[f(t)\right]$$

Substitute $t^2 \cos(3t) u(t)$ for $f(t)$.

$$F(s) = \mathrm{L}\left[t^2 \cos(3t) u(t)\right] \quad \text{...... (1)}$$

Step 3 of 5

Now consider the frequency differentiation using equation 15.34 in the textbook.

$$\mathrm{L}\left[t^n g(t)\right] = (-1)^n \frac{d^n G(s)}{ds^n}$$

Thus, by using the above frequency differentiation property,

$$\begin{aligned}
\mathrm{L}\left[t^2 \cos(3t)\right] &= (-1)^2 \frac{d^2}{ds^2}\left[\frac{s}{s^2+3^2}\right] \qquad \left(\because \mathrm{L}\left[\cos(3t)\right] = \frac{s}{s^2+3^2}\right) \\
&= (-1)^2 \frac{d}{ds}\left[\frac{d}{ds}\left(\frac{s}{s^2+9}\right)\right] \\
&= (-1)^2 \frac{d}{ds}\left[\frac{1.(s^2+9) - 2s.(s)}{(s^2+9)^2}\right] \qquad (\because \text{Using quotient rule of Differentiation}) \\
&= (-1)^2 \frac{d}{ds}\left[\frac{s^2+9-2s^2}{(s^2+9)^2}\right]
\end{aligned}$$

Step 4 of 5

$$L\left[t^2\cos(3t)\right]=(-1)^2\frac{d}{ds}\left[\frac{9-s^2}{\left(s^2+9\right)^2}\right]$$

Apply quotient rule of differentiation to the above function.

$$L\left[t^2\cos(3t)\right]=(-1)^2\left[\frac{(-2s).\left(s^2+9\right)^2-\left(9-s^2\right).\left(2\left(s^2+9\right).(2s)\right)}{\left(\left(s^2+9\right)^2\right)^2}\right] \quad \left(\because \frac{d}{ds}\left(\left(s^2+9\right)^2\right)=2\left(s^2+9\right).(2s)\right)$$

$$=(1).\left[\frac{\left(s^2+9\right)\left[(-2s)\left(s^2+9\right)-4s\left(9-s^2\right)\right]}{\left(s^2+9\right)^4}\right]$$

$$=\left[\frac{(-2s)\left(s^2+9\right)\left[\left(s^2+9\right)+2\left(9-s^2\right)\right]}{\left(s^2+9\right)^4}\right]$$

$$=\left[\frac{(-2s)\left[s^2+9+18-2s^2\right]}{\left(s^2+9\right)^3}\right]$$

$$L\left[t^2\cos(3t)\right]=\left[\frac{(-2s)\left[-s^2+27\right]}{\left(s^2+9\right)^3}\right]$$

$$L\left[t^2\cos(3t)\right]=\frac{(2s)\left(s^2-27\right)}{\left(s^2+9\right)^3}$$

Step 5 of 5

Substitute $\frac{(2s)\left(s^2-27\right)}{\left(s^2+9\right)^3}$ for $L\left[t^2\cos(3t)\right]$ in equation (1).

$$F(s)=L\left[t^2\cos(3t)u(t)\right]$$

$$=\frac{(2s)\left(s^2-27\right)}{\left(s^2+9\right)^3}$$

Therefore, the required Laplace transform of the given function is $\boxed{F(s)=\frac{2s\left(s^2-27\right)}{\left(s^2+9\right)^3}}$.